The lectin reactivity and lectin-like activity of allergenic pollen extracts

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The binding of soluble components of pollen grains to plant-stigma receptors can be inhibited by concanavalin A. This lectin-like activity of pollen components is important in the genetic control of plant reproduction. Aqueous extracts of allergenic pollens also react with concanavalin A. Agarose gel–diffusion precipitates were used to survey and characterize the ability of allergenic pollen extracts to react with concanavalin A and other lectins. Concanavalin A alone precipitated with extracts of plantain, American beech, white ash, and corn pollens. Surprisingly, extracts of the pollen from certain plants also precipitated when the extracts were diffused against pollen extracts from other plants. Pollen extracts of alfalfa, white ash, American beech, burweed marsh elder, redtop grass, corn, plantain, orchard grass, and aspen reacted with one or more other pollen extracts. Extract precipitin activity was reliably obtained after extracting pollens for 20 min with pH 7.5, 0.05M Tris buffer in 0.2M of saline. Optimal agarose gel conditions for detecting the precipitin reactions were pH 8.5 to 9.0, 75 mM borate buffer made to an ionic strength of 1.5M with NaCl for concanavalin A pollen reactions and 0.015M with NaCl for pollen-pollen reactions. The presence of the borate ion was necessary for optimal detection of the agarose gel precipitates. Studies of the inhibition of the lectin-pollen and pollen-pollen reactions with specific mono and disaccharides revealed many similarities and differences between the two types of reactions. The high concentrations of glycerol used to stabilize pollen extracts also inhibit these reactions. (J ALLERGY CLIN IMMUNOL 73:619, 1984.)

The studies presented in this article were prompted by an awareness of pollen-related mechanisms important in plant reproduction. Pollens are biochemically complex structures. The allergenic components of pollens reside in proteins from the wall of the pollen grain. These proteins are rapidly solubilized by moisture and provide cognitive cues to the stigma that control plant reproduction. The seminal observation that a lectin, concanavalin A, could inhibit the binding of a pollen glycoprotein to the stigma surface[1] suggested an alternative mechanism by which allergenic components of pollens might react with animal tissues. Pollen components with similar lectin activity could become concentrated at mucosal sites such as the intercellular cement between mucosal cells[2] and/or induce a lymphoid cell mitogenic response in immunologically naive subjects.[3] Of equal importance, the capacity of lectins, such as concanavalin A, to bind selectively to certain pollen glycoproteins[4] suggests this biologic activity might be used to biochemically isolate allergenic proteins from pollen extracts. This article summarizes studies of allergenic extracts of pollens that react with concanavalin A and the surprising ability of certain pollen extracts to react with pollen extracts from other plants.

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MATERIAL AND METHODS

Raw dry pollens were purchased (Greer Laboratories, Lenoir, N. C.) and defatted with diethyl ether by use of a Soxhlet extractor before preparing allergenic extracts. The defatted pollens were used to prepare 1/5 w/v extracts. Extracts were prepared by eluting soluble compounds from 0.2 gm of dry pollen per 1 ml of solvent for 20 min at room temperature with 0.05M Tris in 0.2M NaCl adjusted to pH 7.5 with 0.1M NaOH.[4] The pollen was separated from the solvent by centrifugation (500 × g, 20 min), and the supernatant extract was decanted, immediately ultrafiltered

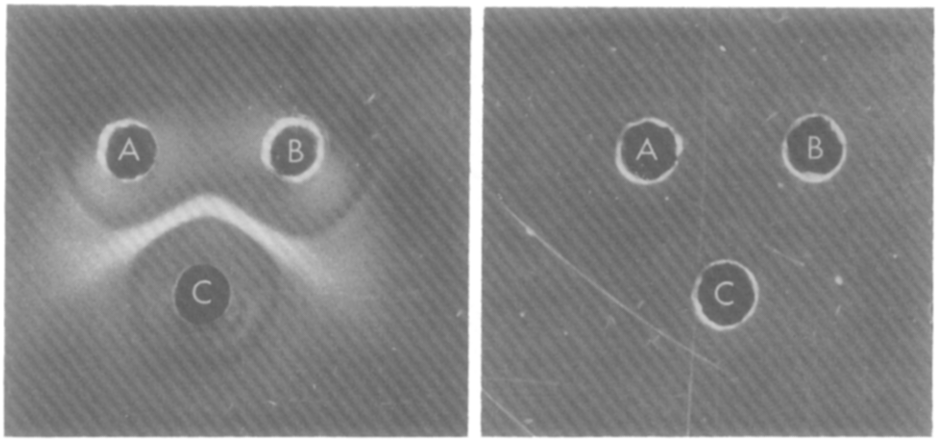

FIG. 1. Left hand panel illustrates precipitin reaction between 0.2% concanavalin A *(c)*, allergenic extracts of corn *(a)*, and American beech *(b)* pollens. Right hand panel illustrates inhibition of both precipitates by 0.05 mM α methyl-D-mannopyranoside in the agarose gel. Note the difference between these unusual precipitates and sharp precipitates usually detected between antigen and antibody.

through a 0.22-micron filter (Millipore Corp., Bedford, Mass.), and stored in the cold at 4° C until used. Freshly prepared extracts from small quantities of pollens were used for most studies, and dry weight measurements of the solute were not done. Selected extracts were lyophilized and reconstituted in order to evaluate the effect of freeze-drying on the activity of the preparation. When the weighed quantity of lyophilized material was corrected for 10% water and the known salt content, the calculated quantity of extracted material varied between 42 and 48 mg of allergenic solute per gram of pollen. These collateral studies suggest the concentration of allergenic solute in the small volumes of $^1/_5$ w/v extracts used in these studies was approximately 9 mg/ml.

Similar representative dry pollen samples were obtained from other manufacturers (Allergy Labs of Ohio, Columbus, Ohio; Hollister-Stier Laboratories, Spokane, Wash.), and small quantities extracted in the same way. In specific experiments, the duration of the extraction time was varied from 5 min to 72 hr. The pH of the extracting solution was adjusted from 2.4 to 11.6 with 0.1 N HCl or 0.1 N NaOH. The ionic strength of the extract solution was varied with NaCl concentration between 0.1M and 0.3M and adjusted to 0.2M at the time of testing.

The pollen extracts used were prepared from redtop grass *(Agrostis stolonifera)*, alfalfa *(Medicago sativa)*, burweed marsh elder *(Iva xanthifolia)*, American beech *(Fagus grandiflolia)*, corn *(Zea mays)*, plantain *(Plantago lanceolata)*, white ash *(Fraxinus americana)*, aspen *(Populus tremuloides)*, orchard grass *(Dactylis glomerata)*, Johnson grass *(Sorghum halepense)*, sweet vernal grass *(Anthoxanthum odoratum)*, rye grass *(Lolium multiform)*, Kentucky blue grass *(Poa pratensis)*, fescue grass *(Festuca elatior)*, timothy grass *(Phleum pratense)*, short ragweed *(Ambrosia artemisifolia)*, and red clover *(Trifolium pratense)*.

Concanavalin A was purchased from Miles-Yeda, Ltd., Elkhart, Ind., and type VI soybean agglutinin, *Tetragonolobus purpuras* lectin, horse gram, peanut agglutinin, and kidney bean lectin were purchased from Sigma Chemical Co., St. Louis, Mo. The concentrations of the lectins were adjusted by dilution based on the manufacturer's estimate of concentration and/or activity. The maximum concentration of concanavalin A used for agarose gel double diffusion was 5.2 mg/ml. Ouchterlony double diffusion[5] in agarose was used to detect precipitin reactions between concanavalin A and other lectins and the allergenic extracts of tree, grass, and weed pollens. A 1% solution of agarose in pH 6.5 75 mM borate buffer made to 0.15M ionic strength with NaCl was used. Double diffusion was also used to assay the precipitin reactions between pollen extracts. Alternative gel-diffusion conditions were evaluated and not found as useful for detecting precipitates. These conditions included the use of agar instead of agarose and the use of Na_2SO_4, Tris chloride, or Na_2PO_4 as the salt or buffer combination in the agarose. Microscope slides (100 by 30 mm) were covered with 2.5 ml of molten agarose, allowed to harden, and diffusion wells were cut with a template (LKB, Rockville, Md.) to yield 6-1.0 mm wells placed circumferentially 1.5 mm from the center well. The wells were filled with 10 μl of each reactant. Gel diffusion was allowed to proceed at 4° C for a maximum of 72 hr and was interpreted as positive or negative on the basis of the presence of a visible precipitate.

The effect of the ionic strength of the diluent used in the agarose gel was evaluated by adding NaCl to yield a total ion concentration varying from 0.015M to 1.5M. In experiments involving ionic strength, the intensity of the precipitate was scored as detectable (+) and strong (++) precipitin reactions. The effect of the diluent pH in the agarose gel was evaluated by adjusting the pH with 0.1 N HCl or 0.1 N NaOH between 3.9 and 11.

Glycerol and selected mono- and disaccharides were used to study the inhibition of the precipitates. Reagent grade glycerol (Baker Chemical Co., Phillipsburg, N. J.), D-glu-

TABLE I. Reaction of allergenic pollen extracts with concanavalin A and other pollen extracts with lectin-like activity*

		Pollen or lectin no. and reaction									
No.	**Reactant**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
1	Concanavalin A										
2	Redtop grass	—									
3	Burweed marsh elder	—	—								
4	American beech	+	+	+							
5	Alfalfa	—	+	+	+						
6	White ash	+	+	+	+	+					
7	Corn	+	+	+	+	—	—				
8	Plantain	+	+	+	—	—	—	—			
9	Aspen	—	+	+	—	—	—	—	—		
10	Orchard grass	—	—	—	+	—	—	—	—	—	

*+ = detectable precipitin reaction; — = no detectable precipitin reaction. The numbers at the top of each column refer to the name and number of the pollen or lectin listed for each row. The concanavalin A concentration was 5.2 mg/ml. The pollen extracts were 1/5 w/v preparations containing approximately 9 mg/ml of allergenic solute. The agarose buffer was made to 0.15M ionic strength with NaCl.

TABLE II. pH Optima for detecting reaction of concanavalin A and pollen extracts*

		Agarose pH							
Reactants		**3.9**	**5.0**	**6.0**	**7.0**	**8.5**	**9.0**	**10**	**11**
Concanavalin A	Plantain	—	—	—	—	+	+	+	+
	American beech	—	—	—	+	+	+	+	+
	White ash	+	+	+	+	+	+	+	+
	Corn	+	+	+	+	+	+	+	+
Alfalfa	White ash	—	—	—	+	+	+	+	—
	American beech	—	—	—	+	+	+	—	—
	Burweed marsh elder	—	—	—	+	+	+	—	—
Redtop	Alfalfa	—	—	—	+	+	+	—	—
	Corn	—	—	—	+	+	+	+	—
	Plantain	—	—	—	+	+	+	+	—
American beech	Redtop	—	—	—	+	+	+	—	—
	Orchard	—	—	—	+	+	+	—	—
	Alfalfa	—	—	—	—	+	+	—	—
Burweed marsh elder	Aspen	—	—	—	—	+	+	+	—
	White ash	—	—	—	—	+	+	+	—
	American beech	—	—	—	—	+	+	—	—

*+ = detectable precipitin reaction; — = no detectable precipitin reaction. The agarose pH was adjusted with weak acid or base as indicated in the text. The agarose ionic strength was 0.15M.

cose, D-mannose, D-fructose, D-ribose, D-xylose, D-arabinose, D-rhamnose, D-galactose, L-fucose (Sigma Chemical Co.), and α-methyl-D-mannopyranoside, N-acetyl-D-galactosamine, and N-acetyl-D-glucosamine (Aldrich Chemical Co., Milwaukee, Wis.) were purchased and used without further modification. The final agarose gel sugar or glycerol concentration was made to 0.05 mM, 0.5 mM, 5 mM, or 50 mM with respect to each inhibiting compound.

RESULTS

Ouchterlony double diffusion of concanavalin A yielded reproducible precipitin lines when it was tested in agarose gel diffusion with four (23%) of the 17 allergenic extracts of pollens. An example of these unusual precipitates (type I reactions) is demonstrated in Fig. 1. Soybean agglutinin, *Tetragonolobus purpuras* lectin, horse gram, peanut agglutinin, and kidney bean lectins did not react with any of the pollen extracts. During these studies, several allergen extracts produced similar agarose gel precipitates with one another (type II reactions). This serendipitous finding was exploited by Ouchterlony-diffusion tests among all the pollen extracts. Table I summarizes the observed reactions between concanavalin A and pol-

TABLE III. The effect of the molarity of sodium chloride in agar gel on the reaction between allergenic extracts of pollens, concanavalin A, and pollen extracts with lectin-like activity

Reactants		Ionic strength (M) of agar gel						
		0.015	0.075	0.150	0.300	0.600	0.900	1.500
Concanavalin A	Corn	—	—	+	+	+	+	++
	Plantain	—	—	+	+	+	+	++
	White ash	—	+	+	++	++	++	++
	American beech	—	—	+	++	++	++	++
Redtop	Alfalfa	++	+	+	—	—	—	—
	Corn	+	+	+	+	+	—	—
	Plantain	++	+	+	+	—	—	—
Alfalfa	White ash	++	++	++	+	+	—	—
	American beech	++	++	++	++	+	+	—
	Burweed marsh elder	++	++	++	+	+	—	—
Burweed marsh elder	Aspen	++	++	+	+	+	+	—
	White ash	++	++	+	+	+	+	—
	American beech	++	++	+	+	+	+	—
American beech	Orchard	++	++	++	++	+	—	—
	Redtop	++	++	++	++	+	+	++
	Alfalfa	++	++	++	++	+	+	—

The intensity of the observed reactions in this experiment was graded as: ++ = strong precipitin reaction; + = detectable precipitin reaction; — = no detectable precipitin reaction. The ionic strength was adjusted with added NaCl (see text). All reactions were at pH 6.5.

len extracts and between the pollen extracts. The allergenic extracts producing precipitin reactions with concanavalin A were from corn, white ash, American beech, and plantain pollens. Allergenic extracts from Johnson grass, sweet vernal grass, rye grass, Kentucky blue grass, fescue grass, redtop grass, orchard grass, timothy grass, alfalfa, aspen, burweed marsh elder, short ragweed, and red clover did not produce precipitin reactions with concanavalin A.

The use of borate buffer and agarose gel was necessary for the optimal detection of these precipitin reactions. An inhibitory effect of sulfate ion was suggested by the detection of the precipitin reactions in agarose gels but not in agar gels or in agarose gels containing added sulfate instead of borate ion. Substitution of phosphate or tris chloride buffers in the agarose gel permitted detection of only a few weak precipitin reactions.

Concanavalin A pollen reactions. The experimental conditions under which concanavalin A-reacting type I compounds could be eluted from pollens were evaluated. Precipitates were detected with pollen extracts eluted for 5 and 20 min and for 72 hr. Variation of the extracting solution pH between 3.9 and 10.3 had no effect on the detection of the precipitates. Variation in the ionic strength of the extracting solution had an effect on the detectability of precipitin reactions. Concanavalin A reacted with soluble components from plantain and American beech pollens eluted in buffers containing 0.1M, 0.2M, and 0.3M ionic strengths. Only the 0.2M buffered extracts prepared from corn and white ash pollens reacted with concanavalin A. Lyophilization of the extracts of the four pollens and reconstitution in appropriate diluent did not alter their ability to react with concanavalin A. These observations suggest the concanavalin A–reacting components extracted from corn, white ash, American beech, and plantain pollens were relatively insensitive to variation in the extraction time, the pH of the buffered extracting solution, and to lyophilization. A 0.2M ion concentration was optimal for eluting reacting components from corn and white ash pollens. The compounds extracted from all four pollens formed visible precipitates when the compounds were diffused against as little as 2 mg/ml concanavalin A in agarose gel.

Optimal experimental conditions for detecting the reaction between concanavalin A and components of the pollen extracts were evaluated (Tables II and III). The precipitin reaction between concanavalin A and corn and white ash pollen extracts remained detectable when the agarose gel pH was varied between 3.9 and 11 (Table II). The detection of the precipitin reaction between concanavalin A and allergenic extracts from plantain and American beech pollens was limited to an alkaline pH range. Variation of the ionic strength of the diluent used in the agarose had a graded effect on the intensity of the visualized precipitates. Among the four pollens reacting with concanavalin A, no precipitates were detected in gels containing 0.015M ionic strength, and only one precipitate was detected in the gels containing 0.075M ionic strength (Table III). Extracts from all four pollens reacted with concanavalin A when the ion concentration in the agarose gel was varied between 0.15M and 1.5M, and sharper precipitin lines were detected with increased ionic strength. A pH between 7.4 and 11 and an ionic strength of 1.5M permitted optimal detection of the reactions between pollen compounds and concanavalin A.

The effect of mono- and disaccharides and glycerol on the reaction between concanavalin A and the four allergenic extracts was evaluated (Table IV). Ouchterlony gel diffusion with 0.2M ionic strength at pH 7.4 revealed inhibition of the precipitin reactions between concanavalin A and all four pollen extracts by the lowest sugar concentration (0.05 mM) of glycerol, D-glucose, D-fructose, D-mannose, and α-methyl-D-mannopyranoside. Ouchterlony double diffusion in the presence of an intermediate concentration (0.50 mM) and higher concentration (5.00 mM) of D-ribose, D-xylose, D-arabinose, L-rhamnose, D-galactose, L-fucose, N-acetyl-D-galactosamine, and N-acetyl-D-glucosamine selectively failed to inhibit various precipitin reactions. The highest concentration (50 mM) of these same sugars uniformly inhibited the precipitin reactions.

Pollen-pollen reactions. Nine (53%) of the 17 allergenic extracts of pollens reacted with other pollen extracts. Table I summarizes the type II reactions observed between the allergenic extracts of these pollens. Allergenic extracts prepared from redtop grass and burweed marsh elder pollens each reacted with the same six extracts among the eight remaining pollens. The extract prepared from American beech pollen reacted with five of the eight remaining pollen extracts. The extracts prepared from alfalfa, corn, plantain, aspen, and orchard grass pollens reacted respectively with four, three, three, two, two and one of the eight remaining extracts. Collectively, 17 out of 36 possible reactions were detected among the nine allergenic pollen extracts.

The importance of variation in the time used to extract soluble compounds from several pollens was evaluated with respect to the detection of the agarose gel precipitin reactions. Extraction of redtop grass pollen for 5 and 20 min yielded detectable precipitin reactions with allergenic extracts from alfalfa, corn, and plantain pollens. Extraction of alfalfa pollen for 5 and 20 min yielded detectable precipitin reactions with allergenic extracts from white ash, American beech, and burweed marsh elder pollens. Extraction of redtop grass pollen for 72 hr yielded detectable precipitin reactions with the alfalfa and plantain pollen extracts. The 72-hour extract of redtop grass pollen did not produce a detectable reaction with the corn pollen extract. A 72-hour extract of the alfalfa pollen did not produce a detectable reaction with white ash, American beech, and burweed marsh elder pollen extracts. These observations suggested short extraction times are important for the detection of the agarose gel–precipitin reactions between allergenic extracts of pollens.

The effect of extract pH on the detection of the pollen-pollen precipitin reactions was evaluated. Variation of the extract pH between 2.4 and 11.6 yielded extracts from redtop grass that reacted with allergenic extracts of alfalfa, plantain, and corn pollens. Variation of the extract pH between 2.4 and 11.6 had no effect on the ability of compounds eluted from alfalfa pollen to react with an allergenic extract from white ash pollen. In contrast, extracts of alfalfa pollen reacted with American beech pollen only when it was prepared in the pH range between 3.8 and 11.6. Extracts of alfalfa pollen at varied pHs reacted with an allergenic extract of burweed marsh elder pollen only when it was prepared in the pH range between 4.1 to 7.7. These observations suggest the pH of the extracting solution is important in the elution of pollen compounds capable of reacting with soluble components of other pollens.

The effect of the molarity of the solution used to extract redtop grass pollen on the detection of precipitin reactions was evaluated with allergenic extracts from corn, alfalfa, and plantain pollens. The redtop grass pollen extracts prepared with 0.1M, 0.2M, and 0.3M ionic strength extract solution reacted with allergenic extracts from alfalfa and plantain pollens. Only the redtop grass extract prepared with 0.2M ionic strength reacted with the allergenic extract of corn pollen. The effect of extract molarity on soluble compounds eluted from alfalfa pollen was evaluated with allergenic extracts from white ash, American beech, and burweed marsh elder pollens. The reaction of the alfalfa extracts were limited to the 0.2M concentration with the exception of a reaction between the 0.1M extract and the allergenic extract from American beech. These observations suggest a 0.2M ionic strength is optimal for the elution of pollen compounds capable of reacting with allergenic extracts of other pollens.

The effect of agarose pH on the detectability of reactions between allergenic extracts from several pollens was examined. The results of these experiments are summarized in Table II, which suggests positive reactions between pollens can be detected over the pH range between 7 to 10.0. This contrasts with the 3.0 to 11 pH range for detecting reactions

TABLE IV. Specific sugar-inhibition concanavalin A precipitin reactions with pollen extracts and of pollen–pollen extract precipitin reactions in agarose gel

	Specific sugars no.*						
	1	2	3	4	5	6	7
	mM of compound inhibiting precipitin reaction						
Concanavalin A/Plantain	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Concanavalin A/American beech	5.00	0.05	0.50	5.00	0.05	0.05	5.00
Concanavalin A/White ash	0.05	0.05	50.00	5.00	5.00	0.05	0.05
Concanavalin A/Corn	0.05	0.05	50.00	5.00	50.00	0.05	5.00
Burweed marsh elder/Corn	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Redtop grass/Aspen	0.05	0.05	0.05	0.50	0.05	0.05	0.05
Burweed marsh elder/Aspen	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Alfalfa/White ash	0.05	0.05	50.00	0.05	0.05	0.05	0.05
Redtop grass/Plantain	0.05	0.05	0.05	5.00	0.05	0.05	0.05
Burweed marsh elder/Plantain	0.05	0.05	0.05	0.05	0.05	5.00	0.50
American beech/Alfalfa	0.05	0.05	50.00	0.05	50.00	0.05	0.05
Burweed marsh elder/White ash	0.05	0.05	0.05	0.05	0.05	5.00	50.00
American beech/White ash	0.05	0.05	50.00	0.05	50.00	0.05	0.05
Redtop grass/Corn	0.05	50.00	0.05	50.00	5.00	50.00	0.05
American beech/Corn	0.05	0.05	50.00	0.05	50.00	5.00	50.00
Burweed marsh elder/Alfalfa	0.05	0.05	50.00	0.05	50.00	50.00	50.00
Redtop grass/White ash	0.05	0.05	5.00	50.00	5.00	50.00	50.00
Burwed marsh elder/American beech	0.05	0.05	50.00	0.05	50.00	50.00	50.00
American beech/Orchard grass	5.00	50.00	50.00	—	—	5.00	—
Redtop grass/Alfalfa	—	—	50.00	—	50.00	50.00	—
Redtop grass/American beech	—	—	50.00	—	50.00	50.00	—

*The specific sugars inhibiting each reaction were: 1 = L-rhamnose; 2 = glycerol; 3 = D-galactose; 4 = N-acetyl-D-galactosamine; 5 = D-ribose; 6 = D-glucose; 7 = D-arabinose; 8 = D-xylose; 9 = D-fructose; 10 = D-mannose; 11 = L-fucose; 12 = α-methyl-D-mannopyranoside; 13 = N-acetyl-D-glucosamine; — = no inhibition observed.

between concanavalin A and pollens. Table II suggests both types of reactions can be most reliably detected with an agarose pH between 8.5 and 9. The effect of varying the ionic strength of the agarose gel on the detectability of the precipitin reactions is summarized in Table III. The pollen-pollen reactions were detected over a wide range from 0.015M to 0.9M, but unlike the reaction between concanavalin A and the allergenic extracts, the reactions between pollen extracts were stronger at lower ionic strengths.

Table IV summarizes studies of the inhibition of the reactions between the allergenic pollen extracts with glycerol and with mono- and disaccharides. These observations are presented in detail because of the potential usefulness of the sugars in the biochemical isolation of pollen components. Seventeen pollen-pollen extract reactions were studied with four concentrations of each inhibiting compound. The concentration in the agarose gel of mono and disaccharides and of glycerol was varied thousandfold between 0.05 and 50 mM. The compounds most capable of inhibiting the precipitin reactions between pollen extracts are presented toward the upper left corner of Table IV. Sugars and pollen-pollen extract reactions exhibiting lesser degrees of inhibition are presented in sequential order toward the lower right corner of Table IV.

Each reaction between the pollen extracts was inhibited by one or more sugars or glycerol. Although minimal concentrations (0.05 mM) of L-rhamnose inhibited a higher number of pollen extract reactions than any other sugar, the ability of glycerol to inhibit the observed precipitin reactions was almost as high. By inspection many similarities in the degree of inhibition by different sugars are evident in Table IV. A remarkable degree of specificity for selective inhibition of different pollen-pollen reactions by certain sugars is also evident. The differences between specific pollen-pollen reactions are as striking as the differences between the concanavalin A reactions with the four pollens presented at the top of Table IV. The information presented in Table IV suggests that the precipitin reactions detected between allergenic extracts of pollens, like the reactions between concanavalin A and allergenic extracts, can be inhibited by glycerol and specific simple sugars.

Specific sugars no.*					
8	**9**	**10**	**11**	**12**	**13**
mM of compound inhibiting precipitin reaction					
0.05	0.05	0.05	0.05	0.05	0.05
0.05	0.05	0.05	0.05	0.05	5.00
5.00	0.05	0.05	0.05	0.05	50.00
50.00	0.05	0.05	50.00	0.05	5.00
0.05	0.05	0.05	50.00	0.05	5.00
0.05	0.05	0.05	0.05	0.05	50.00
0.05	0.05	0.05	50.00	0.05	5.00
0.05	0.05	50.00	0.05	0.05	0.05
0.05	0.05	0.05	0.05	—	50.00
0.05	5.00	5.00	0.05	50.00	5.00
50.00	50.00	50.00	0.05	0.05	0.05
5.00	5.00	5.00	50.00	5.00	50.00
50.00	50.00	50.00	0.05	0.05	0.05
50.00	0.05	0.05	50.00	—	50.00
50.00	50.00	50.00	0.05	—	50.00
50.00	50.00	50.00	—	—	—
50.00	50.00	—	50.00	—	50.00
50.00	50.00	50.00	—	—	—
50.00	—	50.00	—	—	—
50.00	50.00	50.00	—	—	—
—	—	—	—	—	—

DISCUSSION

Four of the 17 (23%) allergenic extracts of common pollens contained soluble compounds capable of reacting with concanavalin A, and nine of the 17 (53%) allergenic extracts contained compounds capable of reacting with components of other allergenic extracts. This suggests the concanavalin A–reacting components of allergenic extracts may be more prevalent than previously appreciated[4] and that certain allergenic extracts contain compounds capable of binding to ligands in other allergenic extracts. Interestingly, the four pollen extracts capable of reacting with concanavalin A were among the nine pollen extracts capable of reacting with one another. The visible precipitation in agarose detected in these studies is consistent with formation of a biologic copolymer by the reactants and implies that two or more binding sites are present on the monomers involved in the polymerization reaction.

With respect to the target ligand, two types of reactions were observed in these studies. The first type of reaction (I) involves a concanavalin A–reactive ligand in the allergenic pollen extract. The second type of reaction (II) involves a lectin-like compound in one allergenic extract and a target ligand in the second allergenic extract. At the present level of development of these observations, it is not possible to know which pollen extract contains the lectin-like component in the type II reaction. The sugar-inhibition studies suggest the type II reaction is complex. Further clarification of the molecular basis for the type I and II reactions must await biochemical isolation of the reacting components of allergic extracts.

Allergenic extracts have been formulated into treatment mixtures by physicians for many years. Agar gel–diffusion methods have been previously used in experimental studies of antigenic components of pollen extracts. The general lack of awareness of these reactions was difficult to understand until optimal conditions for their detection were identified. Gel precipitin reactions between allergenic extracts of pollens and antibody have been traditionally studied in sulfate containing agar gels[6] by use of phosphate-buffered saline.[7] Sulfate and phosphate anions inhibit the precipitin reactions detected in these studies. This suggests the experimental conditions traditionally used to study the precipitation of allergenic extracts with antibody in agar could have compromised the ability to detect these reactions.

Borate buffer is necessary for the optimal formation of these precipitates and implies the participation of glycoproteins in the reaction.[8] This observation is important because borosilicate glass is frequently used to manufacture the vials used to contain allergenic extracts. Mixtures of allergenic extracts for patient use are formulated in borosilicate glass vials. The time-dependent elution of borate ion from the glass vial could contribute to the instability of stored mixtures of allergenic extracts. Sulfate ion inhibits the development of these precipitates. The intensity of the precipitates is decreased in phosphate and tris buffers. Current studies suggest these less intense precipitin reactions can be reliably detected with light scattering nephelometric measurements in phospate-buffered saline in the absence of agarose and borate ion. Complete inhibition of the type II pollen-pollen reactions could improve the stability of allergenic extract mixtures prepared for patient use.

The optimal conditions for (a) preparing the extracts and (b) detecting the precipitin reaction in agarose gel differed for the type I and II reactions. The concanavalin A–reactive type I pollen components eluted best with a pH between 4 and 10, a 0.2M ionic strength, and a 20-minute extraction time. A pH between 8.5 and 11 and 1.5M ionic strength was optimal for detecting the precipitin reactions between concanavalin A and pollen components. Type II pollen-pollen reactive components eluted best with a pH between 4.1 and 7.7, at 0.2M ionic strength, and a 20-minute extraction time. A pH between 8.5 and 9 and 0.015M to 0.150M ionic strength was optimal for detecting the type II pollen-pollen precipitin reactions. These differences and the effect of borate suggest specific ions and hydrophilic interactions as well as hydrophobic forces[9] contribute to the production of the gel precipitin reactions.

Among the four lectins studied, concanavalin A alone reacted with components of allergenic extracts of pollens. Concanavalin A produced type I precipitin reactions with extracts from corn, white ash, American beech, and plantain pollens. This interesting hemagglutinating[9, 10] and mitogenic[11] lectin reacts with naturally occurring structures on pollens as well as the carbohydrate containing structures[12] on neoplastic and normal cells[13-15] previously reported. Although the structural specificity of the reactions with pollen components is not known, it is important to note that the pattern of specific sugar inhibition of concanavalin A precipitation with pollen extracts detected in this study was qualitatively and quantatively the same as the inhibition of concanavalin A agglutination of red blood cells.[9] Collectively, these observations, like the role of borate ion in the observed precipitin reactions, suggest the possible existence of important carbohydrate-containing components of allergenic pollen extracts.

Extracts from American beech, redtop grass, burweed marsh elder, alfalfa, white ash, corn, plantain, aspen, and orchard grass pollens exhibited type II lectin-like precipitin reactions with other pollens. The sugar inhibition studies (Table IV) suggested these reactions are complex. Possible instances of specific inhibition were detected. For example, compare the inhibition of reactions between alfalfa and, respectively, white ash and burweed marsh elder pollen extracts by L-rhamnose, D-galactose, and D-fructose. Glycerol and L-rhamnose were both very efficient inhibitors of all the type II pollen-pollen precipitin reactions detected in these studies. The remaining sugars specifically inhibited a restricted number of type II pollen-pollen precipitin reactions. The effect of glycerine on the many type II pollen-pollen reactions may account for the efficacy of glycerin as a stabilizing agent for allergenic extracts.[16] L-rhamnose inhibited the pollen-pollen reactions as well as glycerin.

These observations have several important fundamental and clinical implications. With respect to allergen isolation, concanavalin A has been used to characterize pollen extracts[17] and separate major allergenic components from Bermuda grass and common weed.[18] Preliminary studies of this possible approach to allergen standardization have been accomplished with a component of an allergenic extract of corn pollen, which was eluted with glucose from a concanavalin A–affinity gel. The isolated component elicits cutaneous reactivity in subjects who are sensitive to allergenic extracts of corn pollen (Klemawesch, S. J., unpublished observations). This suggests the four concanavalin A–reacting allergenic extracts and the lectin-like activity of pollen extracts described in this article provide opportunities for the biochemical isolation of allergen components useful in standardization.

The rapidity with which these components of allergenic extracts can be eluted from pollen grains provides a plausible basis for suggesting their potential clinical importance. A remarkable degree of similarity exists between the few minutes needed for the plant stigmata to hydrate compatible pollen grains,[19] the 20-minute period of time needed to elute the precipitin activity from pollen grains, and the time needed for the moist nasal respiratory mucosa of sensitive subjects to elute and react to pollen allergens. The possibility exists that the same components of allergenic extracts and similar mechanisms are involved in all three reactions.

Another potential source of the clinical importance of these components of allergenic extracts stems from the similarity of the type II pollen-pollen reactions to the reaction between concanavalin A and other naturally occurring substances. Concanavalin A can bind to intercellular substances between respiratory mucosal cells.[2] This suggests the binding of a type II–reacting, concanavalin A–like component of certain pollens to respiratory mucosa can not be excluded. This mechanism could concentrate the pollen compound in the airways and facilitate sensitization.

Finally, the components of allergenic extracts detected in these studies could have an important role in the regulation of the immune response in susceptible individuals. Like concanavalin A, certain pollen components are mitogenic for lymphocytes from agammaglobulinemic subjects, normal individuals, newborn infants, and atopic subjects.[3] In experimental animals, inhalation exposure to concanavalin A and antigen produces experimental hypersensitivity pneumonitis, whereas antigen or concanavalin A exposure alone produce very little change in the lungs.[20] The possibility exists that the lectin-like, type II–reacting component of the allergenic extracts detected in these studies could have a similar effect on the regulation of the immune response. Moreover, recent studies indicate that carbohydrate structures, such as the rutin component of tobacco allergen, can determine the kind of immune response made to the antigen.[21] This suggests that the putative carbohydrate ligands detected in the type I and type II reactions in these studies could contribute to the regulation of allergen sensitization in susceptible individuals. The lectin reactivity and lectin-like activity of allergenic pollen extracts identified in these studies may have important clinical implications in our evolving understanding of pollen-related diseases.

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